

Paper_ED_Cos_06 — Inflationary Fluctuation Spectrum as Substrate-Inherited Sub-Leading Perturbations on the Saturation Regime

Series: Wave-3 Generative Papers (Cosmology Arc — primordial perturbation spectrum from inflation) **Status:** Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime. Direct composition of Cos_01 (saturation-regime inflation framework + Q-C3-2 subleading-order) + Cos_03 + Cos_04 (downstream observable inheritance) + Cos_05 (Route A4 substrate-parameter inheritance) + GW_00 (saddle-Hessian-signature framework for tensor modes) + standard inflationary-perturbation-theory machinery via DCGT continuum bridge. Cos_01 non-load-bearing OPENS (rows 14-16) inherited at upstream status; not load-bearing for Cos_06 M3 verdict. **Date:** 2026-05-17 **Authors:** Paper_087 (13 primitives); Paper_073 (DCGT); Paper_012 (substrate-c); Paper_027 (Newton's G); Paper_ED_SC_4_9 (S1/S2/S3); Paper_ED_GW_00 (M3 — saddle-Hessian-signature framework for tensor modes); **Paper_ED_Cos_01 (M3 unconditional — saturation-regime + Q-C3-2 subleading-order)**; Paper_ED_Cos_03 (M3); Paper_ED_Cos_04 (M3); Paper_ED_Cos_05 (M3); Paper_095 (verdict grammar — §3.2.1 sub-labels). Substrate-research support memos (Route A4 + Q-C3-2 Chain-Class) cross-referenced in §2.

Label-class note: Audit-row sub-labels *substrate-parameter-INHERITED* and *IC-INHERITED* per Paper_095 §3.2.1.

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim closed-proof derivation of standard inflationary-perturbation-theory machinery (Mukhanov-Sasaki equation; quantum-field-theory mode quantization on de Sitter background; slow-roll parameter formalism) from substrate primitives. Standard inflationary perturbation theory is **INHERITED** via DCGT continuum bridge; ED supplies the substrate-graph reading of *what underlies the de Sitter background* (Cos_01 saturation regime) and *how substrate-parameter content enters via Route A4*.
2. It does **not** sit within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). ED is in the substrate-ontology research program. Methodology per Paper_095 (form-FORCED / value-INHERITED).
3. It does **not** introduce new substrate primitives or paper-specific postulates. The inflationary-spectrum content is composition of Cos_01 + Cos_03 + Cos_04 + Cos_05 + Route A4 + standard inflationary-perturbation-theory machinery via DCGT continuum bridge.
4. It does **not** claim numerical-precision derivation of specific inflation parameters ($A_s \approx 2.1 \times 10^{-9}$; $n_s \approx 0.965$; $r < 0.06$; running $\alpha_s \approx -0.005$). These specific numerical values inherit standard CAMB / CLASS Boltzmann-code outputs constrained by Planck + BI-CEP/Keck + future-observatory data; ED supplies the substrate-graph mechanism producing the saturation-regime background, not a substrate-graph rederivation of all spectral parameters.
5. It does **not** address inflation-model-specific predictions (e.g., ϕ^2 chaotic vs Starobinsky vs α -attractor models; specific slow-roll potential shapes). ED's substrate-graph framework is **model-agnostic at substrate level** — the saturation-regime framework (Cos_01) accommodates any standard slow-roll inflation phenomenology; specific inflation-model selection

requires standard astrophysics + observational constraints not derivable substrate-graph.

6. **Verdict M3 via continuum-Friedmann inheritance + standard-inflationary-perturbation-theory inheritance.** Cos_01's non-load-bearing OPENs (row 14 regime duration ~ 60 e-folds; row 15 smooth termination; row 16 V1-transported substrate-graph noise spectral character) are inherited at upstream status as substrate-research-frontier tightening; Cos_06 inherits the standard slow-roll inflation + perturbation-spectrum framework at standard-cosmology level via DCGT bridge, which is M3-equivalent inheritance and supplies Cos_06's net M3 verdict.

Abstract

The inflationary fluctuation spectrum — primordial scalar power $P_\zeta(k) = A_s(k/k_*)^{n_s-1}$ with $n_s \approx 0.965 \pm 0.004$, $A_s \approx 2.1 \times 10^{-9}$, $r < 0.06$ (Planck + BICEP/Keck 2021) — is identified at substrate-graph level as **the inherited sub-leading-order ($O(\varepsilon)$) perturbations around the Cos_01 saturation-regime substrate state, supplied by standard inflationary-perturbation-theory machinery on the substrate-graph-derived de Sitter background.** Sub-leading-order Ψ -fluctuations around uniform- Ψ saturation are admissible per Q-C3-2 (Memo_ED_ChainClass_C3_Audit) and correspond to standard quantum-mode excitations of the inflaton-analog content; **tensor perturbations correspond to substrate-graph saddle-Hessian-signature fluctuations per Paper_ED_GW_00** — the same substrate-graph mechanism that produces post-recombination GW propagation also generates primordial tensor modes (the strongest ED-distinctive cross-arc unification in the Cosmology Arc). Three spectral components: scalar ζ , primordial tensor (GW), running α_s . Standard slow-roll machinery + spectral-parameter calculation inherits via DCGT; Route A4 supplies H_{infl} + present-epoch normalization at substrate-parameter-INHERITED level. **Form-IDENTIFIED** via Cos_01 + GW_00 + standard slow-roll. **Value-INHERITED:** spectral parameters via CAMB/CLASS + Planck/BICEP constraints; H_{infl} , M_P via Route A4 + Paper_027. **IC-INHERITED** from Cos_01 saturation entry (framework-neutral). Substrate-c preserved. **Verdict per Paper_095: M3. F1 (sharpest):** spectral-parameter sensitivity to Route A4 substrate-parameter scaling — i.e., observed A_s and r inconsistent with H_{infl} inherited via $\Theta_{\text{ED}}^{1/2}$ at substrate-parameter-INHERITED tier when standard slow-roll machinery is taken as given — refutes the *Route A4 inheritance channel* specifically (not standard slow-roll inflation). Currently consistent: $n_s = 0.965 \pm 0.004$, $r < 0.06$, $\alpha_s = -0.005 \pm 0.013$ all consistent with standard slow-roll + Route A4 normalization. **Empirical anchors strong:** Planck + BICEP/Keck 2021; future B-mode targets (Simons Observatory, CMB-S4, LiteBIRD); Planck $|f_{\text{NL}}| \sim O(1)$. **Cosmology Arc completion milestone: eight Cosmology Arc papers at M3** (Cos_01, Cos_02, Cos_03, Cos_04, Cos_05, Cos_06, Paper_038_5, Baryogenesis — Paper_038_5 upgraded 2026-05-17 from M2 $\rightarrow$ M3 retroactive via Route A4 substrate-parameter-INHERITED closure; Baryogenesis upgraded 2026-05-17 from M2 $\rightarrow$ M3 via R4 single-template admissibility + IC-supply equivalence per Update_Paper_ED_Baryogenesis_M3_via_R4) structurally spanning primordial perturbations through dark-energy-dominated late universe + Λ -smallness reframing + matter-antimatter asymmetry origin.

§1 Statement of Result

The inflationary fluctuation spectrum is the substrate-graph **sub-leading-order** ($O(\varepsilon)$) **perturbations around the Cos_01 saturation-regime substrate state**, supplied by standard inflationary-perturbation-theory machinery on the substrate-graph-derived de Sitter background:

$$P_\zeta(k) = A_s \left(\frac{k}{k_*} \right)^{n_s-1}, \quad P_T(k) = A_T \left(\frac{k}{k_*} \right)^{n_T}, \quad r = \frac{P_T(k_*)}{P_\zeta(k_*)}$$

with pivot scale $k_* \approx 0.05 \text{ Mpc}^{-1}$, $n_s \approx 0.965$ (slow-roll deviation from scale-invariant $n_s = 1$), $n_T \approx -r/8$ (slow-roll consistency relation), $A_s \approx 2.1 \times 10^{-9}$ (Planck-derived normalization), $r < 0.06$ (Planck + BICEP/Keck 2021 upper limit).

Three spectral components (standard inflationary-perturbation phenomenology, substrate-side reading):

Component	ED substrate-graph mechanism	Observational anchor
Scalar curvature perturbations ζ	Sub-leading $O(\varepsilon)$ Ψ -density fluctuations around uniform- Ψ saturation (Cos_01 Q-C3-2 admissible regime); standard slow-roll inheritance	CMB temperature + polarization (TT, TE, EE; Planck precision); $P(k)$ at large scales (BOSS / DESI)
Tensor perturbations (primordial GW)	Substrate-graph saddle-Hessian-signature fluctuations during inflation per Paper_ED_GW_00 framework; tensor-mode propagation at substrate-c during + post-inflation	CMB B-mode polarization (BICEP/Keck; future Simons Observatory + CMB-S4 + LiteBIRD); stochastic GW background via GW_02 cosmological-source component
Running $\alpha_s = dn_s/d \ln k$	Second-slow-roll-order substrate-graph Ψ -dynamics during inflation regime evolution	CMB + large-scale-structure joint analyses; $\alpha_s \approx -0.005 \pm 0.013$ (Planck precision)

FORM: primordial spectrum = sub-leading perturbations around saturation regime — **IDENTIFIED** via Cos_01 + standard inflationary-perturbation-theory + Route A4 composition. Zero pure-D rows.

VALUE: A_s , n_s , r , α_s all inherit standard inflationary-perturbation-theory + observational constraints via DCGT continuum bridge; H_{infl} + present-epoch normalization inherit Route A4 substrate-parameter-INHERITED scaling at Paper_027-analog tier.

IC: substrate-graph saturation regime at Cos_01 post-BBB ignition (framework-neutral per Memo_ED_CCC_Realizability_SubConstruct: cyclic-CCC vs R3 single-aeon); inflaton-analog substrate-action content via Paper_ED_SC_4_9 + Cos_01 framework.

Verdict: **M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED)** within DCGT A→regime per §4 audit table.

§2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02 (participation), P03 (translation invariance — Noether source), P09 (polarity), P10 (multiple rule-types), P11 (commitment / time arrow), P12 (ED-threshold + stability landscape), P13 (translation arrow).

Upstream papers: - Paper_012 (P-RB-1 substrate-c) - Paper_027 (Newton's G — Planck-scale dimensional rearrangement; $M_P = \sqrt{\hbar c/G}$ appears in inflation-energy-scale parameters) - Paper_073 (DCGT) - Paper_ED_SC_4_9 (saddle classification S1/S2/S3; supplies substrate-action functional structure for Ψ -fluctuation analysis) - Paper_ED_GW_00 (M3 + row 12 partial — supplies substrate-graph saddle-Hessian-signature reconfiguration framework relevant to tensor perturbation modes) - **Paper_ED_Cos_01 (M3 unconditional — supplies saturation-regime substrate-graph framework + Q-C3-2 subleading-order admissible fluctuation discussion + standard slow-roll inflation inheritance)** - Paper_ED_Cos_03 (M3 — observational inheritance for primordial spectrum via $C_\ell^{TT/TE/EE}$ + B-mode polarization targets) - Paper_ED_Cos_04 (M3 — supplies $P(k)$ inheritance chain via $P_{\text{init}}(k) \propto k^{n_s-1} \rightarrow T(k) \rightarrow P(k, a)$ composition) - Paper_ED_Cos_05 (M3 — supplies Route A4 H_0 substrate-parameter-INHERITED scaling for present-epoch normalization) - Update_ED_SC_4x_Arc_M3 - Standard inflationary-perturbation-theory machinery (Mukhanov-Sasaki equation; quantum-field-theory mode quantization on de Sitter background; slow-roll parameter formalism ε, η ; consistency relations $n_T = -r/8$, $n_s - 1 = -6\varepsilon + 2\eta$) — textbook inheritance via DCGT continuum bridge - Standard observational-cosmology machinery (CAMB / CLASS Boltzmann codes for inflationary-spectrum constraints; Planck likelihood; BICEP/Keck likelihood; future Simons Observatory + CMB-S4 + LiteBIRD targets) — textbook inheritance

Substrate-research support: - Memo_ED_RouteA_A4_Audit + Memo_ED_RouteA_MultiRouteConvergen (Route A4 substrate-parameter-INHERITED closure) - Memo_ED_ChainClass_C3_Audit (Q-C3-2 subleading-order admissible fluctuation framework — directly relevant to inflationary fluctuation spectrum as $O(\varepsilon)$ perturbations around saturation) - Memo_ED_Q1Q2_JointClosure_Construct + Memo_ED_ChainClass_C3_Construct

No new substrate primitives. No paper-specific postulates.

§3 Derivation

§3.1 Inflation as saturation regime (Cos_01 inheritance)

Per Paper_ED_Cos_01 (M3 unconditional via Q1A + Q2A + C3 + IC-INHERITED chain): inflation is the substrate-graph **saturation regime** — uniform- Ψ substrate state with V1 + V5 + individuation reactivated above the P12 ED-threshold. Under saturation, substrate-side Noether stress-energy collapses to vacuum-energy form ($T_{\text{sub}}^{\mu\nu} = -g^{\mu\nu} \mathcal{L}_{\text{sub}}^{\text{const}}$); DCGT translates to vacuum-energy continuum stress-energy with $w = -1$; Friedmann recovery gives $a(t) \propto e^{Ht}$ — de Sitter exponential expansion characteristic of inflation.

Inflationary fluctuation spectrum is the sub-leading $O(\varepsilon)$ perturbations around this saturation regime. Per Memo_ED_ChainClass_C3_Audit Q-C3-2: subleading $O(\varepsilon = \ell_{\text{ED}}/R_{\text{cg}})$ fluctuations around uniform- Ψ saturation are admissible at substrate-graph level — these are the

substrate-graph carriers of the inflationary quantum fluctuations that standard inflation theory treats as quantum mode-excitations of the inflaton field on de Sitter background.

Substrate-graph composition: Cos_01 saturation-regime framework + Q-C3-2 subleading-order admissible fluctuations + standard inflationary-perturbation-theory mode quantization (Mukhanov-Sasaki) $\rightarrow$ primordial spectrum.

§3.2 Quantum fluctuations during inflation (subleading-order framework)

Standard inflationary perturbation theory: quantum vacuum fluctuations of the inflaton field ϕ during inflation are stretched to super-horizon scales by the exponential expansion; the fluctuations freeze at horizon crossing ($k = aH$) and re-enter the horizon during post-inflation (RDE $\rightarrow$ MDE) eras, providing the seed for CMB anisotropies + large-scale structure.

ED substrate-graph reading: per Q-C3-2 admissible subleading-order framework, Ψ -fluctuations around the uniform- Ψ saturation regime correspond at substrate-graph level to the standard quantum-mode excitations. The substrate-graph framework supplies the structural identification (subleading- Ψ -fluctuation around uniform substrate state); standard mode-quantization machinery via DCGT continuum bridge supplies the quantitative spectrum.

Mukhanov-Sasaki action for scalar curvature perturbation ζ :

$$S_{\zeta}^{(2)} = \frac{1}{2} \int d^4x a^3 \frac{\dot{\phi}^2}{H^2} \left[\dot{\zeta}^2 - \frac{(\partial\zeta)^2}{a^2} \right]$$

inherits at standard-perturbation-theory level via DCGT. The substrate-graph reading: ζ corresponds to sub-leading Ψ -density fluctuations; the action structure inherits via DCGT continuum bridge from substrate-side Noether content at saturation regime.

§3.3 Scalar perturbation spectrum $P_{\zeta}(k)$

Standard slow-roll result: scalar curvature perturbation power at horizon crossing:

$$P_{\zeta}(k) = \frac{1}{8\pi^2\varepsilon} \frac{H^2}{M_P^2} \Big|_{k=aH}$$

with $\varepsilon = (M_P^2/2)(V'/V)^2$ (first slow-roll parameter). At standard slow-roll: $A_s \equiv P_{\zeta}(k_*) \approx 2.1 \times 10^{-9}$ from Planck-derived normalization.

Spectral tilt:

$$n_s - 1 = -6\varepsilon + 2\eta, \quad \eta = M_P^2(V''/V)$$

with observed $n_s = 0.965 \pm 0.004$ requiring $\varepsilon \sim 0.005 + \eta \sim 0.01$ (standard slow-roll values for plausible inflation models).

ED substrate-graph composition: M_P via Paper_027 (Planck mass from substrate constants); H_{infl} via Cos_01 saturation-regime framework (substrate-parameter-INHERITED via Route A4 at Paper_027-analog tier for inflation-epoch H); ε, η inherit standard slow-roll machinery via DCGT

continuum bridge. Specific $A_s + n_s + \alpha_s$ values inherit standard Planck + BICEP/Keck observational constraints.

§3.4 Tensor perturbation spectrum $P_T(k)$

Standard slow-roll result: primordial tensor (gravitational-wave) perturbation power:

$$P_T(k) = \frac{2}{\pi^2} \frac{H^2}{M_P^2} \Big|_{k=aH}, \quad n_T = -2\varepsilon$$

Tensor-to-scalar ratio:

$$r = \frac{P_T}{P_\zeta} = 16\varepsilon$$

at first slow-roll order. Observed $r < 0.06$ (Planck + BICEP/Keck 2021) constrains $\varepsilon < 0.004$ at slow-roll level.

ED substrate-graph reading: primordial tensor perturbations correspond to substrate-graph **saddle-Hessian-signature fluctuations during inflation** per Paper_ED_GW_00 framework — the same substrate-graph mechanism that produces post-recombination GW propagation also generates inflationary tensor modes, but with the inflation-epoch boundary condition (sub-horizon quantum fluctuations during de Sitter expansion become super-horizon classical perturbations at horizon crossing). Tensor-mode propagation at substrate-c (Paper_012 + Paper_089) during + post-inflation is consistent with GW170817 multi-messenger constraint inherited from Paper_ED_GW_00 §3.8.

Future B-mode polarization observations (Simons Observatory, CMB-S4, LiteBIRD) provide tightening constraints on r ; substrate-graph framework predicts standard slow-roll consistency relation $n_T = -r/8$ via inheritance.

§3.5 Slow-roll parameters and spectral tilt

Standard slow-roll formalism: hierarchy of slow-roll parameters

$$\varepsilon = \frac{M_P^2}{2} \left(\frac{V'}{V} \right)^2, \quad \eta = M_P^2 \frac{V''}{V}, \quad \xi^2 = M_P^4 \frac{V'V'''}{V^2}, \dots$$

with $n_s - 1 = -6\varepsilon + 2\eta$, $r = 16\varepsilon$, and running $\alpha_s = dn_s/d\ln k = 16\varepsilon\eta - 24\varepsilon^2 - 2\xi^2$ at second-slow-roll order.

ED substrate-graph reading: slow-roll parameters correspond to the **rate of substrate-action density change** as the substrate evolves through the late-inflation phase. Per Cos_01 §3.5 (smooth-termination OPEN) + §3.3 (diffusion-production imbalance), the substrate-graph dynamics during inflation termination involve substrate-action density evolution that maps onto standard slow-roll potential-evolution at continuum level. Specific slow-roll-parameter values inherit standard inflation-model + observational constraints.

Observed: $\alpha_s = -0.005 \pm 0.013$ (Planck precision) — consistent with zero; future precision improvements (Simons Observatory, CMB-S4) will refine the running constraint and potentially distinguish ED-saturation-regime inheritance from alternative inflation frameworks.

§3.6 Sensitivity to Θ_{ED} or τ_{V_5}

Substrate-parameter Θ_{ED} enters inflationary observables through Route A4's $H_0 \propto \Theta_{\text{ED}}^{1/2}$ and inflation-epoch H_{infl} via Cos_01 saturation-regime framework:

- $A_s \propto H_{\text{infl}}^2/(M_P^2 \varepsilon)$: substrate-parameter content enters via $H_{\text{infl}} + M_P$ both at substrate-parameter-INHERITED tier; A_s value inherits substrate-parameter scaling
- n_s : governed by slow-roll dynamics; substrate-parameter content enters subleading via inflation-model-dependent potential shape; observed $n_s \approx 0.965$ inherits standard inflation-model fits
- r : $\propto \varepsilon \propto H_{\text{infl}}^2/M_P^2$ at leading slow-roll order; substrate-parameter content enters via Route A4 inflation-epoch scaling
- α_s : second-order in slow-roll; substrate-parameter content enters at subleading-subleading level

Substrate-parameter τ_{V_5} enters via Route A1/A2 alternative (multi-route convergence option (ii)) at same substrate-parameter-INHERITED level; inflation-epoch V5 cross-chain content affects fluctuation-spectrum sub-leading content (Cos_01 audit row 16: V1-transported substrate-graph noise spectral character OPEN — substrate-research-frontier tightening).

Tightening via RA-OPEN-4a/4b/4c-explicit substrate-research-frontier, not blocking for Cos_06's M3 verdict.

§3.7 Comparison to Λ CDM inflationary spectrum

Element	Λ CDM standard slow-roll inflation	ED substrate-graph reading	Status
Inflation-epoch background	de Sitter expansion driven by inflaton-field slow-roll	Cos_01 saturation-regime substrate-graph framework + Q1A + Q2A + C3 + IC-INHERITED chain (M3)	D-via-I via continuum-Friedmann inheritance
Quantum vacuum fluctuations during inflation	Standard QFT mode quantization on de Sitter	Sub-leading $O(\varepsilon)$ Ψ -fluctuations per Q-C3-2 admissible framework + standard mode-quantization via DCGT	D-via-I + standard inheritance
Scalar curvature spectrum $P_\zeta(k)$	Standard slow-roll: $P_\zeta = H^2/(8\pi^2 \varepsilon M_P^2)$	Same; substrate-graph H_{infl} via Cos_01 + Route A4 + M_P via Paper_027	D-via-I

Element	Λ CDM standard slow-roll inflation	ED substrate-graph reading	Status
Tensor primordial spectrum $P_T(k)$	Standard: $P_T = 2H^2/(\pi^2 M_P^2)$	Substrate-graph saddle-Hessian- signature fluctuations per Paper_ED_GW_00 framework + standard inheritance	D-via-I + substrate-graph reading
Spectral tilt $n_s - 1 = -6\varepsilon + 2\eta$	Standard slow-roll relation	Same; slow-roll parameters inherit standard inflation phenomenology	D-via-I
Tensor-to-scalar ratio $r = 16\varepsilon$	Standard slow-roll relation	Same	D-via-I
Consistency relation $n_T = -r/8$	Standard	Same; substrate-graph framework inherits	D-via-I
Running α_s	Second slow-roll order	Same	D-via-I
Substrate-parameter Θ_{ED} sensitivity	None	$H_{\text{infl}} \propto \Theta_{\text{ED}}^{1/2}$ via Route A4 $\rightarrow$ spectrum normalization inherits	Distinguishing content
Inflation model-selection (ϕ^2 vs Starobinsky vs α -attractor)	Standard astrophysics + observational constraints	ED substrate-graph framework is model-agnostic at substrate level; saturation-regime accommodates any standard slow-roll model	Inherited (substrate-graph framework supplies saturation-regime; specific model inherits)

§4 Audit Table

#	Step	Label	Notes
1	P02, P03, P09, P10, P11, P12, P13	P	Paper_087.

#	Step	Label	Notes
2	Upstream inheritance block	I	Paper_012; Paper_027; Paper_073; Paper_ED_SC_4_9; Paper_ED_GW_00 (M3 + row 12 partial); Paper_ED_Cos_01 (M3 unconditional); Paper_ED_Cos_03 (M3); Paper_ED_Cos_04 (M3); Paper_ED_Cos_05 (M3); Up- date_ED_SC_4x_Arc_M3.
3	Standard inflationary-perturbation-theory machinery (Mukhanov-Sasaki equation; QFT mode quantization on de Sitter; slow-roll formalism; consistency relations)	I	Textbook inheritance via DCGT continuum bridge.
4	Standard observational-cosmology machinery (CAMB / CLASS for inflationary-spectrum constraints; Planck + BICEP/Keck likelihoods)	I	Textbook inheritance for specific A_s, n_s, r, α_s observational constraints.
5	Inflation as saturation-regime substrate-graph background	D-via-I	§3.1. Inherits Cos_01 (M3 unconditional via Q1A + Q2A + C3 + IC-INHERITED chain).
6	Sub-leading $O(\varepsilon)$ Ψ -fluctuations around uniform- Ψ saturation correspond to standard inflationary quantum fluctuations	D-via-I	§3.2. Composition of Cos_01 saturation framework + Memo_ED_ChainClass_C3_Audit Q-C3-2 admissible subleading-order fluctuations.

#	Step	Label	Notes
7	Scalar curvature spectrum $P_\zeta(k) = A_s(k/k_*)^{n_s-1}$ with $A_s \propto H_{\text{infl}}^2/(M_P^2 \varepsilon)$	D-via-I	§3.3. Composition of standard slow-roll machinery + Cos_01 H_{infl} + Paper_027 M_P .
8	Tensor primordial spectrum $P_T(k) = A_T(k/k_*)^{n_T}$ corresponds to substrate-graph saddle-Hessian-signature fluctuations during inflation	D-via-I	§3.4. Composition of Paper_ED_GW_00 saddle-Hessian-signature framework + Cos_01 + standard tensor-mode quantization.
9	Tensor-to-scalar ratio $r = 16\varepsilon$ at first slow-roll order; consistency relation $n_T = -r/8$	D-via-I	§3.4. Standard slow-roll inheritance.
10	Spectral tilt $n_s - 1 = -6\varepsilon + 2\eta$ via slow-roll parameters ε, η	D-via-I	§3.5. Standard slow-roll formalism + Cos_01 substrate-graph framework.
11	Running $\alpha_s = dn_s/d \ln k$ at second slow-roll order	D-via-I	§3.5. Standard second-slow-roll formalism.
12	Specific $A_s \approx 2.1 \times 10^{-9}$, $n_s \approx 0.965$, $r < 0.06$, $\alpha_s \approx -0.005$ values	I	§3.3-§3.5. Inherits Planck + BICEP/Keck observational constraints.
13	Substrate-parameter Θ_{ED} sensitivity: $A_s \propto H_{\text{infl}}^2$ inherits Route A4 substrate-parameter-INHERITED scaling	D-via-I	§3.6. Composition of Route A4 + standard slow-roll inflation.
14	Substrate-c bound preserved throughout inflation + post-inflation tensor-mode propagation	D-via-I	§3.4. Paper_012 + Paper_089.

#	Step	Label	Notes
15	$H_{\text{infl}} + H_0$ at Route A4 substrate-parameter-INHERITED level	I	Sub-type per Paper_095 §3.2.1: <i>substrate-parameter-INHERITED</i> . §3.3, §3.6. Paper_027-analog tier per Memo_RouteA_MultiRouteConvergence option (ii).
16	Saturation-regime substrate-graph initial conditions (post-BBB ignition, framework-neutral per Q-C3-3)	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §2.IC. Cos_01 framework + Memo_ED_CCC_Realizability_Subcyclic-CCC vs R3 framework-neutrality.
17	Inflaton-analog substrate-action functional content via Paper_ED_SC_4_9 + Cos_01 framework	I	Sub-type per Paper_095 §3.2.1: <i>IC-INHERITED</i> . §2.IC. Substrate-graph framework supplies what underlies the inflaton-analog; specific potential shape inherits inflation-model selection at standard astrophysics level.
18	Cos_01 non-load-bearing OPENs (row 14 ~60 e-folds duration; row 15 smooth termination; row 16 spectral character) inherited at upstream substrate-research-frontier status	OPEN (non-load-bearing)	§1 + §3.5. Substrate-research-frontier tightening; not load-bearing for Cos_06 M3 inheritance verdict.

Verdict M3
 (form-IDENTIFIED +
 value-INHERITED +
 IC-INHERITED)
 within DCGT
 A→regime

A→position

Per Paper_095 §3.3
 line 77. Five-anchor
 verdict-sync: status /
 abstract / §1 / this
 row / §6 all M3.

Audit summary: zero pure-D rows; **ten substantive D-via-I rows** (5-11, 13, 14); four I rows (2, 3, 4, 12); one substrate-parameter-INHERITED row (15); two IC-INHERITED rows (16, 17); one non-load-bearing OPEN (18 — Cos_01 upstream substrate-research-frontier items); one P row (1); one A→position verdict row (19). **No load-bearing OPEN items.** Cos_01 non-load-bearing OPENs inherited at upstream status; not load-bearing for Cos_06 M3 verdict via standard inflationary-perturbation-theory continuum-bridge inheritance. No new substrate primitives; no paper-specific postulates.

§5 Falsification Criteria

- **F1 (sharpest — Route-A4-specific):** A_s and r jointly inconsistent with H_{infl} inherited via Route A4 substrate-parameter scaling $H_{\text{infl}}^2 \propto \Theta_{\text{ED}}$ at Paper_027-analog tier, *when standard slow-roll machinery is taken as given*. This is an ED-distinctive falsifier of the Route A4 inheritance channel — standard slow-roll inflation itself is not the target of F1 (ED inherits it); the Route A4 substrate-parameter normalization channel is. Refutes §3.3 + §3.4 + §3.6 + audit row 15. Currently consistent at Planck + BICEP/Keck precision.
- **F2 (Route-A4-specific):** Detection of $r > 0.06$ at next-generation precision (Simons Observatory, CMB-S4, LiteBIRD) accompanied by a measured H_{infl} inconsistent with Route A4 substrate-parameter scaling at substrate-parameter-INHERITED tier — i.e., the substrate-graph saddle-Hessian-signature framework can accommodate any r value (standard slow-roll predicts $r = 16\varepsilon$ for any ε), but the *Route A4 inflation-epoch normalization* must be consistent with A_s . A detected r requiring H_{infl} in tension with Route A4 substrate-parameter inheritance refutes that channel specifically. Refutes §3.4 + §3.6 + audit rows 8, 15.
- **F3:** Significant violation of slow-roll consistency relation $n_T = -r/8$ in joint $r + n_T$ measurements from future B-mode observations. Refutes §3.4 + audit row 9. (Tests inherited standard-inflation framework; not ED-distinctive.)
- **F4:** Detection of non-Gaussianity (f_{NL}) significantly inconsistent with substrate-graph Ψ -fluctuation framework + standard slow-roll prediction $f_{\text{NL}}^{\text{local}} \sim O(\varepsilon)$. Refutes §3.2 + audit row 6. Currently consistent: Planck $|f_{\text{NL}}| \sim O(1)$ across local / equilateral / orthogonal templates — one of the stronger empirical constraints on substrate-graph Ψ -fluctuation framework.

§6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program,

Sorkin et al.), not within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

The inflationary fluctuation spectrum in ED is the substrate-graph reading of standard inflationary-perturbation-theory on the Cos_01 saturation-regime substrate-graph background: scalar curvature perturbations correspond to sub-leading $O(\varepsilon)$ Ψ -fluctuations around uniform- Ψ saturation per Q-C3-2 admissible framework; tensor perturbations correspond to substrate-graph saddle-Hessian-signature fluctuations per Paper_ED_GW_00 framework; standard slow-roll machinery + Mukhanov-Sasaki mode quantization inherit via DCGT continuum bridge; substrate-parameter content $H_{\text{infl}} + M_P$ inherit Route A4 substrate-parameter-INHERITED scaling at Paper_027-analog tier. Substrate-c bound preserved throughout inflation + post-inflation tensor-mode propagation; GW170817 multi-messenger constraint inherited from Paper_ED_GW_00 §3.8.

Verdict: M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED) within DCGT A→regime hydrodynamic-window per Paper_095. Ten D-via-I rows; four I rows; one substrate-parameter-INHERITED row (Route A4); two IC-INHERITED rows (Cos_01 saturation IC + substrate-action functional content); zero pure-D rows; no paper-specific postulates; no new substrate primitives; **no load-bearing OPEN items**. One non-load-bearing OPEN (Cos_01 upstream substrate-research-frontier items — ~60 e-folds duration, smooth termination, spectral character; inherited at upstream status as substrate-research-frontier tightening).

Substrate-ontology distinguishing content (the genuine ED contributions): (i) **Substrate-graph saddle-Hessian-signature framework for primordial tensor perturbations** per Paper_ED_GW_00 — primordial tensor modes and post-recombination GW propagation share the *same* substrate-graph mechanism. This is the cleanest cross-arc unification in the Cosmology Arc: a single substrate-graph saddle-Hessian-signature framework spans inflation-epoch tensor generation through Hubble-epoch GW propagation. (ii) **Route A4 substrate-parameter Θ_{ED} scaling channel for primordial-spectrum normalization** $A_s \propto H_{\text{infl}}^2 \propto \Theta_{\text{ED}}$ — supplies ED’s structural link to spectral-normalization observational targets. Future B-mode polarization observations + non-Gaussianity measurements + slow-roll consistency tests provide ED-vs-alternative-inflation discriminator channels via these two pathways at substrate-parameter-INHERITED level. **Empirical-anchor strength** at the Planck + BICEP/Keck 2021 level (and future Simons Observatory / CMB-S4 / LiteBIRD precision) is exceptionally strong; Planck $|f_{\text{NL}}| \sim O(1)$ constraints across local / equilateral / orthogonal templates are one of the sharpest existing tests of substrate-graph Ψ -fluctuation framework.

Cosmology Arc completion milestone. Cos_01 (Inflation, M3 unconditional) + Cos_02 (BBN, M3) + Cos_03 (CMB Acoustic, M3) + Cos_04 (Structure Formation, M3) + Cos_05 (Dark Energy, M3 unconditional) + **Cos_06 (this paper, M3)** + Paper_038_5 (Λ -smallness reframing, **M3 retroactive — upgraded 2026-05-17 via Route A4 substrate-parameter-INHERITED closure** per Memo_ED_RouteA_MultiRouteConvergence_Audit option (ii)) + **Paper_ED_Baryogenesis (M3 — upgraded 2026-05-17 from M2 via R4 single-template admissibility + IC-supply equivalence per Update_Paper_ED_Baryogenesis_M3_via_R4)** form the substantively-complete **Cosmology Arc at eight papers M3**, spanning primordial perturbations → nucleosynthesis → CMB acoustic peaks → linear structure formation → dark-energy-dominated late universe + Λ -smallness reframing + matter-antimatter asymmetry origin. All eight carry consistent label-grounding via Paper_095 §3.2.1; Cos_02 / Cos_03 / Cos_04 also carry the architectural-division Preamble item w.r.t. Dyn_02 M2 inheritance.

Cross-arc consequences. Upstream, Cos_06 inherits unchanged from Cos_01 (M3 uncondi-

tional — non-load-bearing OPENs at upstream status), Cos_02 / Cos_03 / Cos_04 / Cos_05 (M3), GW_00 (M3 — saddle-Hessian-signature framework for primordial tensor modes), and the ED-SC 4.x arc-wide M3 picture. Downstream / cross-arc: *strengthens* CMB B-mode polarization observational targets (BICEP, Simons Observatory, CMB-S4, LiteBIRD) via the substrate-graph reading of primordial tensor-mode generation; *strengthens* non-Gaussianity observational targets ($f_{\text{NL}}^{\text{local}}$, $f_{\text{NL}}^{\text{equil}}$) via the substrate-graph Ψ -fluctuation framework; *enables* the slow-roll consistency relation $n_T = -r/8$ as a substrate-graph-supported observational target; *strengthens* SCBU BH-horizon + cosmic-decoupling projections via the cross-arc primordial-tensor-mode anchor at substrate-parameter-INHERITED level via Route A4.

End Paper_ED_Cos_06.